AIM : Digital Signature

IImplement digital signature algorithms such as RSA-based signatures, and verify the integrity and authenticity of digitally signed messages.

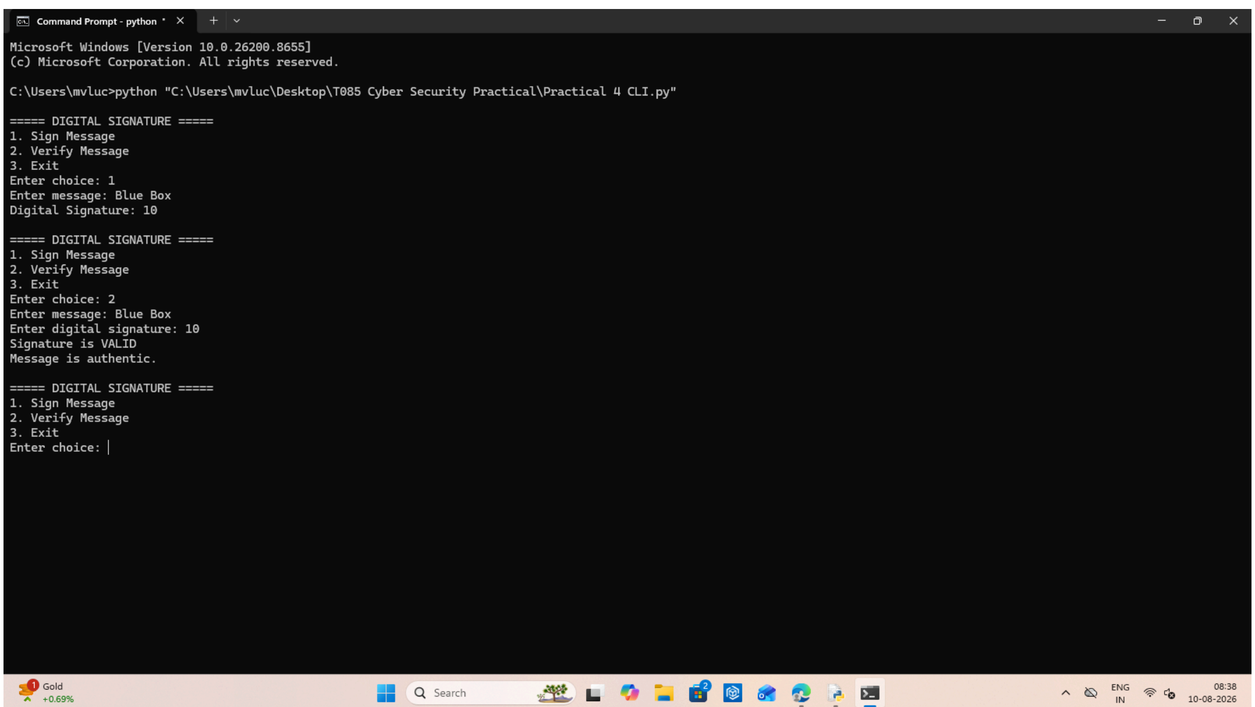
CLI Version :

Code :

```
import hashlib

# RSA values
n = 3233
e = 17
d = 2753

while True:
    print("\n===== DIGITAL SIGNATURE =====")
    print("1. Sign Message")
    print("2. Verify Message")
    print("3. Exit")

    choice = input("Enter choice: ")

    if choice == "1":
        message = input("Enter message: ")

        # Hash message
        h = int(hashlib.sha256(message.encode()).hexdigest(), 16)

        # Generate signature
        signature = pow(h, d, n)

        print("Digital Signature:", signature)

    elif choice == "2":
        message = input("Enter message: ")
        signature = int(input("Enter digital signature: "))

        # Hash message
```

```
        h = int(hashlib.sha256(message.encode()).hexdigest(), 16)

        # Verify signature
        check = pow(signature, e, n)

        if check == h % n:
            print("Signature is VALID")
            print("Message is authentic.")
        else:
            print("Signature is INVALID")
            print("Message may have been changed.")

    elif choice == "3":
        print("Exiting...")
        break

    else:
        print("Invalid choice!")
```

Output :



```
Microsoft Windows [Version 10.0.26200.8655]
(c) Microsoft Corporation. All rights reserved.

C:\Users\mvluc>python "C:\Users\mvluc\Desktop\T085 Cyber Security Practical\Practical 4 CLI.py"

===== DIGITAL SIGNATURE =====
1. Sign Message
2. Verify Message
3. Exit
Enter choice: 1
Enter message: Blue Box
Digital Signature: 10

===== DIGITAL SIGNATURE =====
1. Sign Message
2. Verify Message
3. Exit
Enter choice: 2
Enter message: Blue Box
Enter digital signature: 10
Signature is VALID
Message is authentic.

===== DIGITAL SIGNATURE =====
1. Sign Message
2. Verify Message
3. Exit
Enter choice:
```

GUI Version :

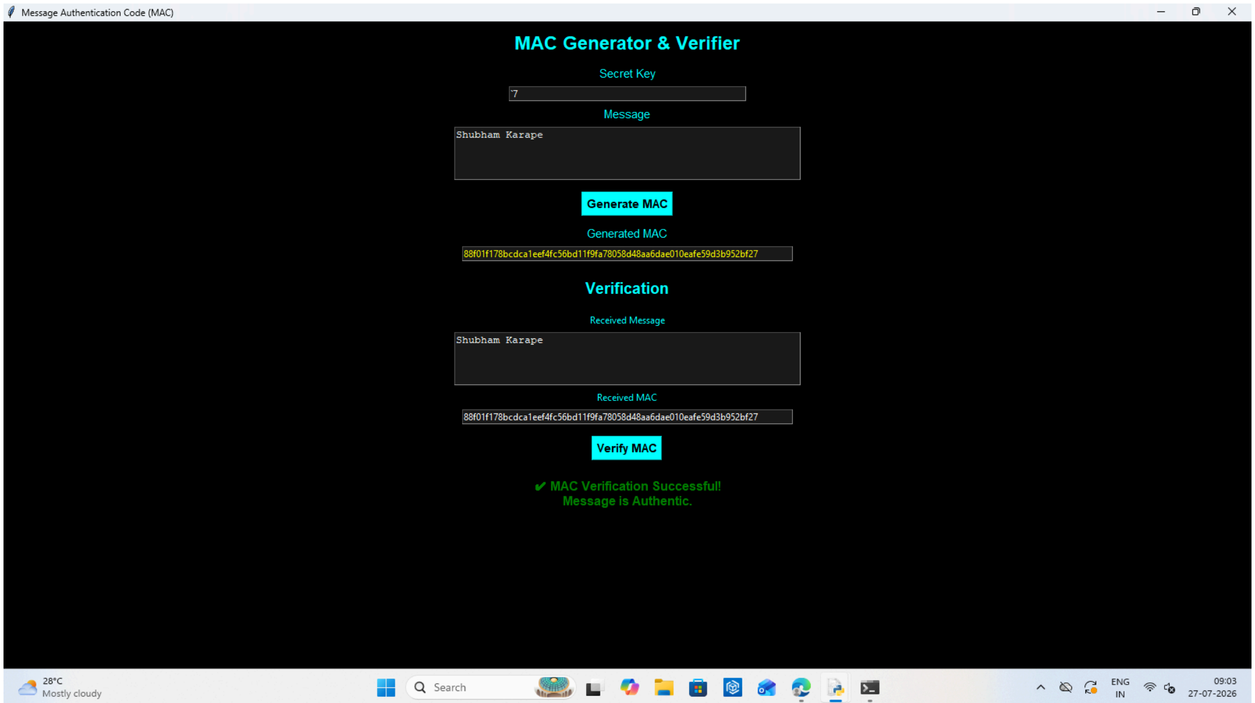
Code :

```
import tkinter as tk
import hashlib

n = 3233
e = 17
d = 2753


def sign():
    message = msg.get("1.0", tk.END).strip()

    h = int(hashlib.sha256(message.encode()).hexdigest(), 16)
    signature = pow(h, d, n)

    sig.delete("1.0", tk.END)
    sig.insert("1.0", str(signature))


def verify():
    message = msg.get("1.0", tk.END).strip()
    signature = int(sig.get("1.0", tk.END).strip())

    h = int(hashlib.sha256(message.encode()).hexdigest(), 16)
    check = pow(signature, e, n)

    if check == h % n:
        result.config(text="✓ Signature Valid", fg="#00ffff")
    else:
        result.config(text="✗ Signature Invalid", fg="red")


# Window
root = tk.Tk()
root.title("RSA Digital Signature")
root.attributes("-fullscreen", True)
root.configure(bg="black")

# Title
tk.Label(
    root,
    text="RSA DIGITAL SIGNATURE",
    font=("Arial", 28, "bold"),
```

```
    fg="#00ffff",
    bg="black"
).pack(pady=40)

# Message
tk.Label(
    root,
    text="Enter Message",
    font=("Arial", 16),
    fg="#00ffff",
    bg="black"
).pack()

msg = tk.Text(
    root,
    height=6,
    width=80,
    bg="#111111",
    fg="#00ffff",
    insertbackground="#00ffff",
    font=("Arial", 14)
)
msg.pack(pady=15)

# Signature
tk.Label(
    root,
    text="Digital Signature",
    font=("Arial", 16),
    fg="#00ffff",
    bg="black"
).pack()

sig = tk.Text(
    root,
    height=3,
    width=80,
    bg="#111111",
    fg="#00ffff",
    insertbackground="#00ffff",
    font=("Arial", 14)
)
sig.pack(pady=15)

# Buttons
tk.Button(
    root,
```

```
    text="SIGN",
    command=sign,
    width=15,
    bg="#00ffff",
    fg="black",
    font=("Arial", 12, "bold")
).pack(pady=10)

tk.Button(
    root,
    text="VERIFY",
    command=verify,
    width=15,
    bg="#00ffff",
    fg="black",
    font=("Arial", 12, "bold")
).pack(pady=10)

# Result
result = tk.Label(
    root,
    text="",
    font=("Arial", 18, "bold"),
    bg="black"
)
result.pack(pady=20)

root.mainloop()
```

Output :


Message Authentication Code (MAC)
MAC Generator & Verifier
Secret Key
7
Message
Shubham Karape
Generate MAC
Generated MAC
88f01f178bcdca1eef4fc56bd11f9fa78058d48aa6dae010eafe59d3b952bf27
Verification
Received Message
Shubham Karape
Received MAC
88f01f178bcdca1eef4fc56bd11f9fa78058d48aa6dae010eafe59d3b952bf27
Verify MAC
✔ MAC Verification Successful!
Message is Authentic.
28°C
Mostly cloudy
Search
ENG
IN
09:03
27-07-2026